# Atom DTU NBIoT2

SKU: K059-B




















## Description

**Atom DTU NBIoT2** is a programmable wireless data transmission unit (DTU) suitable for global Cat-NB frequency bands. It features a built-in SIM7028 communication module, uses serial communication (controlled via AT commands), and integrates an SMA external antenna interface, significantly improving communication quality and signal stability. The SIM7028 module covers most Cat-NB frequency bands, ensuring global applicability. Unlike general DTUs that only support data passthrough, the Atom DTU series adopts a more open architecture design. The AtomS3-Lite controller allows users to modify the program according to actual business needs, and the device

provides multiple interfaces (RS485, I2C, custom interfaces) for user expansion, facilitating quick integration of sensors and actuators. The built-in rail clamping structure allows it to be seamlessly embedded into various industrial control environments. **Atom DTU NBIoT2** is ideal for low-latency, low-throughput applications such as remote control, asset tracking, remote monitoring, telemedicine, and shared bikes.

## Features

- Global version/multi-band support
- Strong signal access capability
- AT command control
- SIM card type: Micro SIM
- External antenna: SMA antenna
- Serial communication: UART 115200bps
- Data transmission (kbps): 127 (DL)/159 (UL)
- Supports multiple network protocols
- Multiple certifications for radio, environmental, and carrier standards
- Multiple power supply methods

## Includes

- 1 x AtomS3-Lite
- 1 x Atom DTU NBIoT2
- 1 x Rubber antenna (2.5dBi, total length 110mm, SMA inner pin)
- 1 x SMA antenna red cap
- 1 x M2 hex wrench
- 1 x M2x16 screw
- 1 x HT3.96-4P terminal

## Applications

- Asset tracking
- Remote monitoring/control
- Shared bikes

## Specifications

| Specification | Parameters |
| --- | --- |
| Communication module | SIM7028 |
| Supported Cat-NB bands | B1/B2/B3/B4/B5/B8/B12/B13/B14/B17/B18/B19/B20/B25/B26/B28/B66/B70/B85 |
| Network protocols | TCP/UDP/HTTP/TLS*/DTLS*/DNS/NTP/PING/LWM2M/COAP*/MQTT/MQTTS/SSL |
| Carrier certifications | Deutsche Telekom/Vodafone/Telefonica/China Telecom/China Mobile/China Unicom |
| Communication method | UART 115200bps |
| Antenna | Rubber antenna @2.5dBi, total length 110mm, SMA inner pin |
| Power supply methods | RS485 port (HT396 interface 9-24V)<br>USB 5V |
| Grove interface load capacity | DC 5V/0.9A |
| Operating temperature | 0 ~ 40°C |
| Product Size | 65.3 x 28.7 x 24mm |
| Product weight | 24.6g |
| Package Size | 168 x 117 x 19.7mm |
| Gross Weight | 38.7g |

## RS485 Communication Test

- Test conditions: AB with 120Ω terminal resistor

| Communication distance | Data rate |
| --- | --- |
| 5 meters | Maximum data rate 750Kbps, normal transmission, 0% packet loss, 0% error rate |
| 20 meters | Maximum data rate 750Kbps, normal transmission, 0% packet loss, 0% error rate |
| 50 meters | Maximum data rate 750Kbps, normal transmission, 0% packet loss, 2% error rate |
| 100 meters | Maximum data rate 512Kbps, normal transmission, 0% packet loss, 0% error rate |

## Power Consumption Test

| Parameter category | Specifications |
| --- | --- |
| Standby current (with Atom-Lite host) | DC 9V@36.00mA<br>DC 12V@28.95mA<br>DC 24V@17.00mA |
| Operating current (with Atom-Lite host) | DC 9V@58.81.00mA<br>DC 12V@48.52mA<br>DC 24V@25.00mA |

# Certifications

## Radio and Communication Equipment Certifications

- **SRRC**, **NAL**, **CE-RED**, **GCF**, **CCC**

## Safety and Environmental Certifications

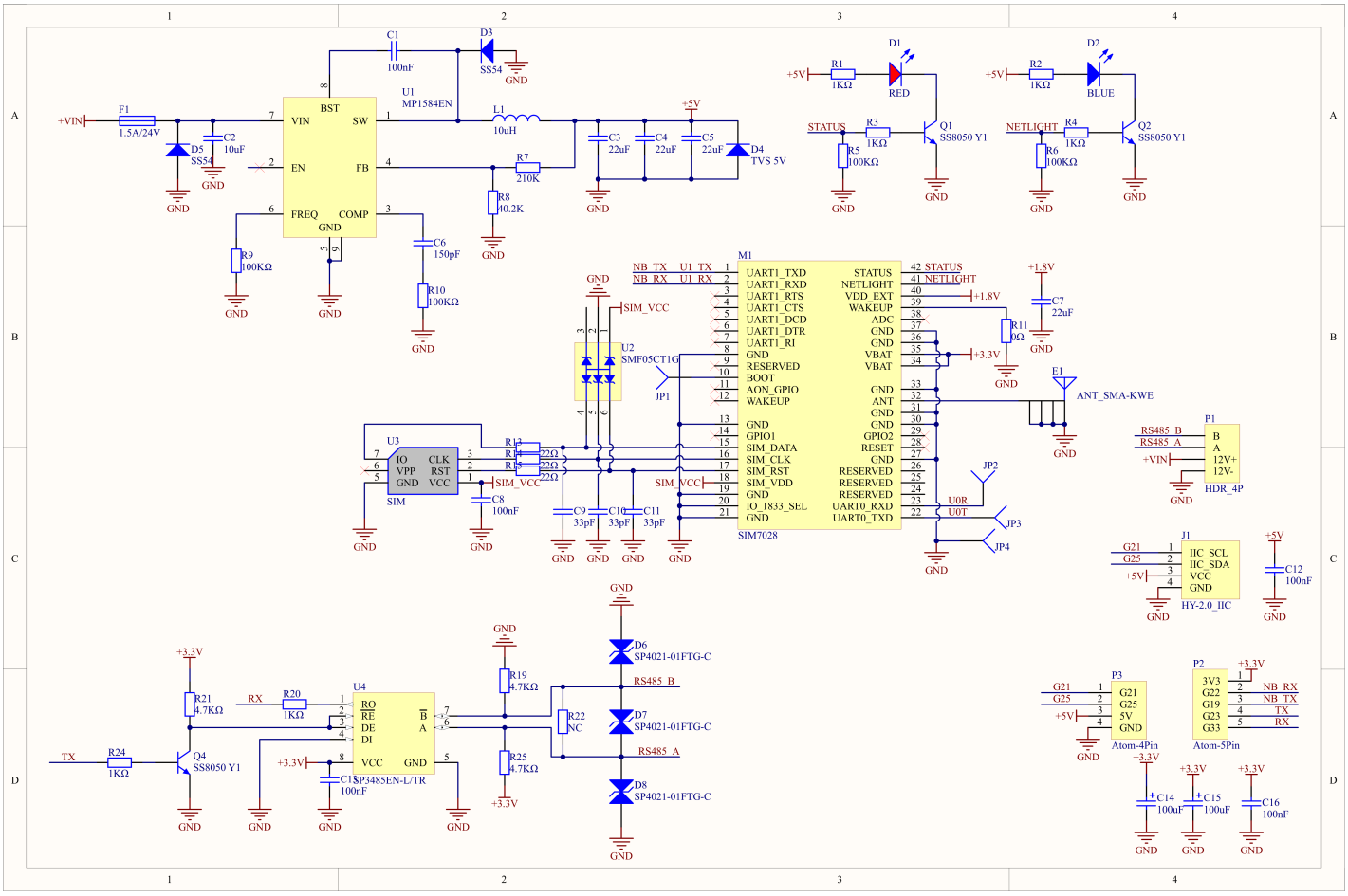
- **CCC**, **RoHS**, **REACH**

## Carrier Certifications

- Deutsche Telekom, Vodafone, Telefonica, China Telecom, China Mobile, China Unicom

# Schematics

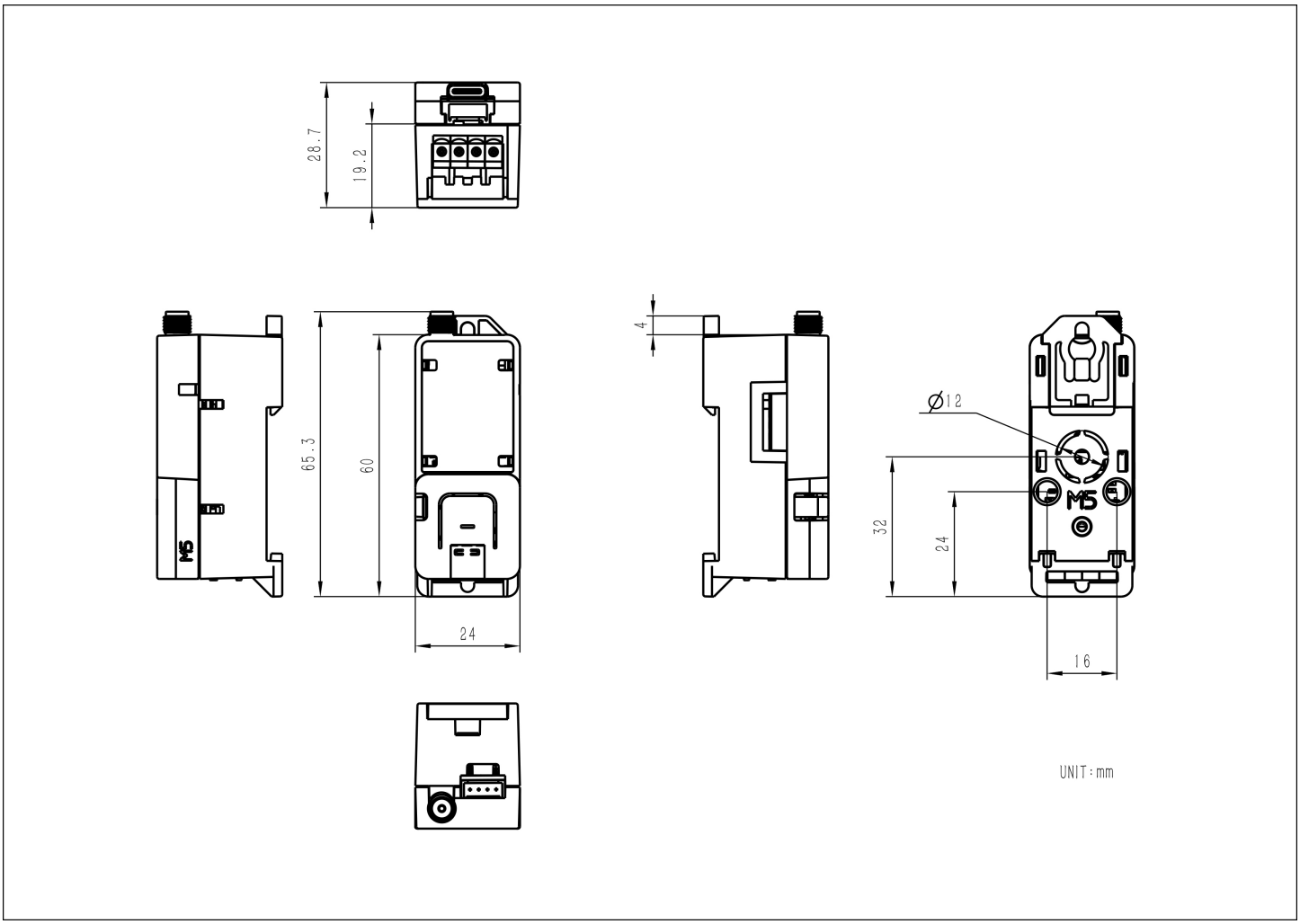
- Atom DTU NBIoT2 Schematics PDF




# PinMap

| PIN | LEFT | RIGHT | PIN |
|---|---|---|---|
| | | 1 | 3V3 |
| PORT.A | 2 | 3 | NBIoT2_UART_RX |
| PORT.A | 4 | 5 | NBIoT2_UART_TX |
| 5V | 6 | 7 | RS485_TX |
| GND | 8 | 9 | RS485_RX |

# Model Size



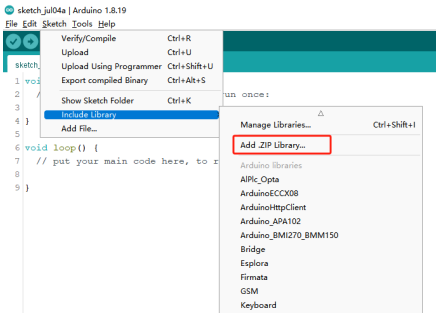

## Datasheets

- SIM7028_Specification
- SIM7028 Series_Network Searching_Application Note_V1.00
- SIM7028 Series_Low Power Mode_Application Note_V1.01
- SIM7028_NCC_Certificate_20230421
- SIM7028 Series_TCPIP_Application Note_V1.04
- SIM7028 Series_SSL_Application Note_V1.00
- SIM7028 Series_MQTT(S)_Application Note_V1.03
- SIM7028 Series_LwM2M_Application Note_V1.02
- SIM7028 Series_HTTP(S)_Application Note_V1.04
- SIM7028 Series_FOTA_Application Note_V1.00
- SIM7028 Series_COAP_Application Note_V1.00
- NB-EVB_User_Guide_Manual_V1.00
- RSU062-EE-EUT Photo-SIM7028
- RSU062-E6-CE EMC Test Report
- RSU062-E5-3GPP (LTE Band 5) Test Report

## Softwares

### Arduino

- Atom DTU NBIoT2 Arduino Library
- Download the library file and manually add it to the Arduino library folder path, or import it directly through the Arduino IDE.




## UiFlow2

- Atom DTU NBIoT2 UiFlow2 Docs

## PlatformIO

- Atom DTU NBIoT2 Project Demo for PlatformIO
- Specify the dependency library version in the `platformio.ini` file.

```
lib_deps =
    m5stack/M5Unified@^0.1.13
    m5stack/M5AtomS3@^1.0.0
    fastled/FastLED@^3.6.0
    knolleary/PubSubClient@^2.8
    vshymanskyy/StreamDebugger@^1.0.1
    https://github.com/m5stack/TinyGSM/archive/refs/tags/v0.12.0-patch-1.zip
    arduino-libraries/ArduinoHttpClient@^0.6.0
    bakercp/CRC32@^2.0.0
```

## Protocol

- SIM7028 Series_AT Command Manual_V1.00

# Video

719dbbd4009f13a6893266f53b301f71.mp4

# Product Comparison

| Product | SKU | Module | Cat-NB Bands | Data Transmission Rate | Communication Protocols |
|---|---|---|---|---|---|
| Atom DTU-NB-IoT | K059 | SIM7020G | B1/B2/B3/B4/B5/B8/B12/B13/B17/B18/B19/B20/B25/B26 /B28/B66/B70/B71/B85 | 126 (DL) / 150 (UL) | TCP/UDP/HTTP/HTTPS/TLS*/DTLS*/DNS/NTP/PING/LWM2M/ COAP*/MQTT/MQTTS/SSL |
| Atom DTU-NB-IoT2 | K059-B | SIM7028 | B1/B2/B3/B4/B5/B8/B12/B13/B14/B17/B18/B19/B20/B25/ B26/B28/B66/B70/B85 | 127 (DL) / 159 (UL) | TCP/UDP/HTTP/HTTPS/TLS*/DTLS*/DNS/NTP/PING/LWM2M/ COAP*/MQTT/MQTTS/SSL |

**Summary of Differences**

The main differences between SIM7028 and SIM7020G are in band support and data transmission rates. SIM7028 supports more

bands and slightly higher data transmission rates, making it suitable for more markets and higher bandwidth applications.

SIM7020G, while supporting fewer bands, still covers most markets and is suitable for general IoT applications. Both modules are similar in network protocols and functional features, with extensive international and carrier certifications.